Aditya Jha

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EDUCATION

Georgia Institute of Technology *May 2029*

Bachelor of Science in Computer Science, *Threads: Artificial Intelligence, Systems & Architecture*

Current Coursework: Linear Algebra, Object Oriented Programming, Physics (mechanics)

TECHNICAL SKILLS

Languages: Python, Java, JavaScript
Machine Learning: TensorFlow, Keras, CNNs, Transfer Learning, Computer Vision, Grad-CAM
Software & Tools: React, Spring Boot, PostgreSQL, Git, GitHub, REST APIs

EXPERIENCE

Undergraduate Researcher at Georgia Institute of Technology | Fedele Lab *Sep 2026 – Present*

- Developing a real-time computer vision pipeline that uses YOLO-based object tracking, nonlinear dimensionality reduction (LLE), and Stable Diffusion to transform tracked motion into generative visual outputs
- Improving tracking stability by fine-tuning YOLO and experimenting with motion-feature extraction to better capture movement speed and direction
- Exploring new representations and processing methods for live visual motion, with an emphasis on real-time inference and human-AI creative applications

First Author & Researcher — IEEE MIT URTC | Advised by Dr. Ajay Bansal, ASU *Apr 2025 – Oct 2025*

Impact: *Built an accurate, interpretable, and validated Alzheimer's classification system*

- Developed a ResNet-style CNN in Keras to classify four Alzheimer's stages from 2D brain MRIs using OASIS and ADNI datasets
- Improved accuracy from 78% to 94% using residual blocks, regularization, dropout, and augmentation
- Integrated Grad-CAM explainability to identify artifact-driven overfitting, guide model redesign, and validate attention on clinically relevant brain regions with professional medical review
- Presented and published at the IEEE MIT Undergraduate Research Technology Conference (URTC)

Founder — FruitShare.org *Mar 2024 – Aug 2026*

Impact: *Grew organization through technology and outreach, converting food surplus into community donations*

- Founded a nonprofit that has recovered and donated 5,000+ lbs of surplus fruit from 50+ homeowners across 10 cities, leading volunteer operations, community outreach, and growth strategy
- Engineered fruitshare.org using React, Spring Boot, and PostgreSQL to manage sign-ups, donation tracking, and fruit-recovery workflows for volunteers and donors via REST APIs, automated emails, and a CRM
- Incorporated community feedback into the website's design and features to ensure accessibility across all ages

Independent Researcher — Vanderbilt Young Scientist Journal *May 2024 – Aug 2024*

Impact: *Demonstrated efficient, high-accuracy deep-learning architectures for malaria detection*

- Trained MobileNetV2 and EfficientNetB0 in Keras on 15,031 blood-smear images to classify parasitized vs. uninfected cells
- Achieved 96% accuracy with MobileNetV2, focusing on reducing CPU and memory usage
- Published research in the Vanderbilt Young Scientist Journal

PROGRAMMING PROJECTS

1st Place Winner, HackGT 13 SpaceXAI Track | React, MapLibre, FastAPI, LLM APIs, Tiger Data *Sep 2026*

- Built Ripple, an infrastructure resilience simulator that models cascading impacts of MARTA disruptions across Atlanta communities and essential services and develops optimal infrastructure strengthening plans. Placed 1st out of 52 teams.
- Designed a geospatial data pipeline with MARTA GTFS, U.S Census, OpenStreetmap, and CMS to model city infrastructure, calculate the complex impact of disruptions, and facilitate an algorithm that calculates optimal spots to build new facilities
- Integrated Grok and Gemini into the deterministic simulation engine to translate and visualize natural-language disaster scenarios as structured infrastructure disruptions

CAMPUS INVOLVEMENT

AI Safety Initiative at Georgia Tech - *AI Safety Fundamentals Fellowship*
Photography @ GT - *Member*

Interests: Photography, basketball, competitive Rubik's Cube solving, weightlifting, sketching, reading
Spoken Languages: English (Native), Hindi (Native)